

Requirements

Functional Requirements Specification

Functional Requirements for Admin (Seller):

- **Authentication & Account Management**
 - Admin can register a new account.
 - Admin can log in using email and password.
 - Admin can request a password reset.
 - The system sends an OTP via email to verify the password reset.
 - Admin enters a valid OTP to create a new password.
 - Admin can log out of the system.
- **Category Management**
 - Admin can create a new category.
 - The system displays a list of all categories.
 - Admin can filter categories.
 - Admin can delete a category.
 - Admin can delete multiple categories at once.
 - Admin can edit a category.
- **Pet Management**
 - Admin can create a new pet with details: name, breed, age, gender, price, category, description, image.
 - The system displays the list of pets.
 - Admin filter pets.
 - Admin can delete a pet.
 - Admin can delete multiple pets at once.
 - Admin can edit a pet.
- **Admin Approval**
 - Current admins can approve newly registered admin accounts.
 - The system displays a list of all admins (approved or pending approval).

- **Order Management**

- The system displays a list of orders.
- Admin can update order status.
- Admin can view order details.

Functional Requirements for Client (Buyer):

- **Product Browsing**

- Users can view the list of pets by category.
- Users can view detailed information of a pet (images, price, description, features, condition, etc.).

- **Searching & Filtering**

- Users can search for pets by name or category.
- Users can filter pets.

- **Shopping Cart**

- Users can add pets to the shopping cart.
- Users can view the entire shopping cart.
- Users can update the quantity of pets in the shopping cart.
- Users can remove pets from the shopping cart.

- **Checkout & Payment**

- Users can pay for all items in the shopping cart.
- The system supports payment via VNPay, MoMo, and ZaloPay.
- The system records the order and sends a notification to the customer.

- **Order Tracking**

- Users can view the list of their placed orders.
- Users can view detailed information of each order.

Non-Functional Requirements Specification

- **Performance Requirements**

- The main pages (home page, product page, shopping cart) must load within ≤ 5 seconds on an average computer configuration.
- The system must run stably when at least 10 users are using it concurrently in the demo environment.

- **Security Requirements**

- User passwords must be encrypted before storing in the database.
- The system must validate and block abnormal input fields to reduce the risk.

- **Reliability Requirements**

- The system must not crash when performing basic operations (adding to cart, logging in, viewing products).
- Data must be safely stored in the database.

- **Usability Requirements**

- The interface must be easy to understand and operate for general users.
- Colors and layout must be clear and suitable for an e-commerce website.

- **Maintainability Requirements**

- Code must be organized in a clear directory structure (Models, Views, Controllers, or equivalent).
- Basic comments must be provided to improve readability and maintainability.